Variable Selection for Wasserstein Robust Hypothesis Testing

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n_k available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^{n_k}$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \delta_{x_i^{(k)}}$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \le \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \le d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Finite reformulation

Assume $n_0, n_1 \ge 1$ and $\rho_0, \rho_1 > 0$. Being real-valued and convex, ℓ is continuous. For $S \subseteq [D]$ let $V(S) = \min_{f \in \mathcal{F}_S} R(f)$; the minimum is attained by Lemma 2.1. Put $\sigma_0 = -1$ and $\sigma_1 = 1$, so that the class-k loss in Eq. (1) is $\ell(\sigma_k f(X))$.

Notation. Pool the samples as $x_1, \ldots, x_N$, where $N = n_0 + n_1$, $x_r = x_r^{(0)}$ for $r \le n_0$ and $x_{n_0+j} = x_j^{(1)}$; they serve as transport targets. For a source $x_i^{(k)}$ and a target x_r, let $\Delta_{ir}^{(k)} \in \mathbb{R}_+^D$ have entries $\Delta_{ir}^{(k)}[m] = |x_i^{(k)}[m] - x_r[m]|$, so that $\|x_i^{(k)}[S] - x_r[S]\|_1 = \mathbf{1}_S^\top \Delta_{ir}^{(k)}$. A plan is a pair $\gamma = (\gamma^{(0)}, \gamma^{(1)})$ with $\gamma^{(k)} \in \mathbb{R}_+^{n_k \times N}$; the entry $\gamma_{ir}^{(k)}$ is the mass moved from $x_i^{(k)}$ to the point that agrees with x_r on S and with $x_i^{(k)}$ off S, at cost $\mathbf{1}_S^\top \Delta_{ir}^{(k)}$. A plan has target masses and coordinate costs

$$p_k^\gamma(r) = \sum_{i=1}^{n_k} \gamma_{ir}^{(k)}, \qquad c_k(\gamma) = \sum_{i=1}^{n_k} \sum_{r=1}^{N} \gamma_{ir}^{(k)} \Delta_{ir}^{(k)} \in \mathbb{R}_+^D, \qquad k = 0, 1.$$

Let Π be the set of plans with $\sum_r \gamma_{ir}^{(k)} = 1/n_k$ for all k and i, and let

$$\Gamma_S = \left\{ \gamma \in \Pi : \mathbf{1}_S^\top c_k(\gamma) \le \rho_k, \ k = 0, 1 \right\}.$$

For $a, b \ge 0$ let $\psi(a, b) = \inf_{t \in \mathbb{R}} \{a\ell(-t) + b\ell(t)\}$, and for every plan γ define

$$H(\gamma) = \sum_{r=1}^{N} \psi\left(p_0^\gamma(r), p_1^\gamma(r)\right) = \inf_{g \in \mathbb{R}^N} \sum_{k=0}^{1} \sum_{i=1}^{n_k} \sum_{r=1}^{N} \gamma_{ir}^{(k)} \ell(\sigma_k g_r). \tag{3}$$

As an infimum of linear functions with nonnegative coefficients, ψ is concave, positively homogeneous, upper semicontinuous and nonnegative. Since p_k^γ is linear in γ, H has the same properties. For the hinge loss $\ell(t) = (1+t)_+$, where $t_+ = \max\{t, 0\}$, one has $\psi(a, b) = 2\min\{a, b\}$.

Lemma 2.1 (Finite reformulation)**.** *For every $S \subseteq [D]$,*

$$V(S) = \max_{\gamma \in \Gamma_S} H(\gamma), \tag{4}$$

and

$$\begin{aligned} V(S) = \min_{g,a,\lambda} \quad & \sum_{k=0}^{1} \Big(\frac{1}{n_k} \sum_{i=1}^{n_k} a_i^{(k)} + \rho_k \lambda_k \Big) \\ \textit{subject to} \quad & \ell(\sigma_k g_r) \le a_i^{(k)} + \lambda_k \mathbf{1}_S^\top \Delta_{ir}^{(k)}, \qquad k = 0, 1,\ i \in [n_k],\ r \in [N], \\ & g \in \mathbb{R}^N, \quad a^{(k)} \in \mathbb{R}^{n_k}, \quad \lambda \in \mathbb{R}_+^2. \end{aligned} \tag{5}$$

Every minimizer of Eq. (5) extends to a detector $f \in \mathcal{F}_S$ with $R(f) = V(S)$.

Eq. (4) is the least favorable distribution program of [8, Lemma 2] (see also [5]) with the transport cost restricted to S: least favorable distributions move mass only along selected coordinates, onto the S-projections of the pooled samples. Eq. (5) is its dual. There, g_r is the detector score at $x_r[S]$, $a_i^{(k)}$ bounds the loss that the adversary extracts from the source $x_i^{(k)}$, and λ_k prices the budget ρ_k. Appendix A proves the lemma and constructs the detector. We use two consequences below. First, V is nonincreasing, since $\mathcal{F}_S \subseteq \mathcal{F}_{S'}$ for $S \subseteq S'$. Second, $0 \le V \le V(\emptyset) = 2$: the bound $\psi(a, b) \le (a+b)\ell(0)$ gives $H \le 2$ on Π, and for $S = \emptyset$ the plan that moves all mass onto one target is feasible with $H = \psi(1, 1) = 2$, because $\ell(-t) + \ell(t) \ge 2\ell(0)$.

3 A convex selection function

Lemma 2.1 gives one program for each subset. The subset enters only through the budgets $\mathbf{1}_S^\top c_k(\gamma) \le \rho_k$, which are bilinear in $(\mathbf{1}_S, \gamma)$. We replace them by a priced form. A price $\mu_m \ge 0$ removes the transport cost on coordinate m from the budget, and the objective charges for it only on selected coordinates. Let $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$.

Theorem 3.1 (Convex selection function)**.** *For $z \in \mathbb{R}_+^D$ define*

$$F(z) = \max_{\gamma \in \Pi,\ \mu \in \mathbb{R}_+^D} \frac{H(\gamma)}{1 + \mu^\top z} \quad \textit{subject to} \quad \sum_{m=1}^{D} \big(c_k(\gamma)[m] - \rho_k \mu_m \big)_+ \le \rho_k, \quad k = 0, 1. \tag{6}$$

(a) F is convex and nonincreasing, and $0 \le F \le 2$.

(b) $F(\mathbf{1}_S) = V(S)$ for every $S \subseteq [D]$. Hence $\min_{|S| \le d} V(S) = \min_{z \in \mathcal{Z}_d} F(z)$.

(c) Let (γ, μ) be feasible in Eq. (6) and $\bar{z} \in \mathbb{R}_+^D$. With $v = H(\gamma)/(1 + \mu^\top \bar{z})$ and $\bar{\mu} = \mu/(1 + \mu^\top \bar{z})$,

$$F(z) \ge v\big(1 - \bar{\mu}^\top (z - \bar{z})\big) \qquad \textit{for all } z \in \mathbb{R}_+^D, \tag{7}$$

with equality at $z = \bar{z}$ if (γ, μ) is optimal at $\bar{z}$.

Proof. The maximum in Eq. (6) exists: the feasible set is closed, the objective is upper semicontinuous, and replacing μ_m by $\min\{\mu_m, \kappa_m\}$ with $\kappa_m = \max_{k,i,r} \Delta_{ir}^{(k)}[m]/\rho_k$ keeps feasibility (as $c_k(\gamma)[m] \le \rho_k \kappa_m$ on Π) and does not decrease the objective. So the maximization may be restricted to a compact set.

(a) For fixed feasible (γ, μ), the map $z \mapsto H(\gamma)/(1 + \mu^\top z)$ composes the convex decreasing function $u \mapsto H(\gamma)/(1+u)$ on $[0, \infty)$ with the nondecreasing linear map $z \mapsto \mu^\top z$, so it is convex and nonincreasing on $\mathbb{R}_+^D$. Pointwise maxima keep both properties, and $0 \le H \le 2$ on Π.

(b) Fix S. Let $\gamma^\star$ attain Eq. (4), and set $\mu_m^\star = 0$ for $m \in S$ and $\mu_m^\star = \max_k c_k(\gamma^\star)[m]/\rho_k$ for $m \notin S$. Then $\sum_m (c_k(\gamma^\star)[m] - \rho_k \mu_m^\star)_+ = \mathbf{1}_S^\top c_k(\gamma^\star) \le \rho_k$ and $(\mu^\star)^\top \mathbf{1}_S = 0$, so $F(\mathbf{1}_S) \ge H(\gamma^\star) = V(S)$.

Conversely, let (γ, μ) be feasible and $\theta = 1/(1 + \mu^\top \mathbf{1}_S)$. Let $\iota \in \Pi$ leave every sample in place: $\iota_{ir}^{(k)} = 1/n_k$ if r is the pooled index of $x_i^{(k)}$, and $\iota_{ir}^{(k)} = 0$ otherwise. Then $\gamma' = \theta\gamma + (1-\theta)\iota \in \Pi$, $c_k(\iota) = 0$, and

$$\mathbf{1}_S^\top c_k(\gamma') = \theta \sum_{m \in S} c_k(\gamma)[m] \le \theta \sum_{m \in S} \big[(c_k(\gamma)[m] - \rho_k \mu_m)_+ + \rho_k \mu_m\big] \le \theta \rho_k (1 + \mu^\top \mathbf{1}_S) = \rho_k.$$

Hence $\gamma' \in \Gamma_S$, and since H is concave and nonnegative,

$$V(S) \ge H(\gamma') \ge \theta H(\gamma) + (1-\theta) H(\iota) \ge \frac{H(\gamma)}{1 + \mu^\top \mathbf{1}_S}.$$

Maximizing over (γ, μ) gives $V(S) \ge F(\mathbf{1}_S)$. The last claim follows from the correspondence $S \leftrightarrow \mathbf{1}_S$.

(c) Convexity of $u \mapsto 1/(1+u)$ gives $1/(1+u) \ge 1/(1+\bar{u}) - (u - \bar{u})/(1+\bar{u})^2$ for $u, \bar{u} \ge 0$. Take $u = \mu^\top z$ and $\bar{u} = \mu^\top \bar{z}$ and multiply by $H(\gamma) \ge 0$. The left side becomes $H(\gamma)/(1 + \mu^\top z) \le F(z)$ and the right side becomes that of Eq. (7). If (γ, μ) is optimal at $\bar{z}$, then $v = F(\bar{z})$. □

What is relaxed. Eq. (6) keeps the plan set Π and the objective H of Lemma 2.1; only the budgets are priced. At $z = \mathbf{1}_S$ the proof shows that pricing is free off S, which removes those coordinates from the budgets, and that pricing on S gains nothing over mixing the plan with ι. At fractional z, coordinate m is charged in proportion to z_m. Thus F is exact on $\{0,1\}^D$, and its minimum over the convex hull of $\mathcal{Z}_d$ is a lower bound on the optimal value.

Remark 3.2 (Saddle-point form)**.** For $h \ge 0$ and $u > 0$, $h/u = \max_{s \ge 0}\{2\sqrt{sh} - su\}$, attained at $s = h/u^2$. Hence

$$F(z) = \max\Big\{2\sqrt{sH(\gamma)} - s - s\mu^\top z : \ (\gamma, \mu) \text{ feasible in Eq. (6)}, \ s \ge 0\Big\}, \tag{8}$$

a maximum of functions affine in z over a set that does not depend on z. This is the structure of [1, Theorem 2.5] on which outer approximation rests. The tangent Eq. (7) is the affine piece with $s = v/(1 + \mu^\top \bar{z})$.

4 Outer approximation

Evaluating F. For a fixed z, Eq. (6) is a concave–convex fractional program. The next proposition gives its Charnes–Cooper form [3], which the algorithm solves, and its dual, which explains the coupling of z with the detector variables.

Proposition 4.1 (Evaluation of F)**.** *Let $z \in [0,1]^D$.*

(a) $F(z)$ is the optimal value of

$$\begin{aligned} \max_{\hat{\gamma}, \hat{\mu}, \beta, \alpha \ge 0} \quad & H(\hat{\gamma}) \\ \text{subject to} \quad & \textstyle\sum_r \hat{\gamma}_{ir}^{(k)} = \alpha / n_k, && k = 0,1,\ i \in [n_k], \\ & c_k(\hat{\gamma})[m] \le \beta_{km} + \rho_k \hat{\mu}_m, \quad \textstyle\sum_m \beta_{km} \le \rho_k \alpha, && k = 0,1,\ m \in [D], \\ & \alpha + \hat{\mu}^\top z = 1. \end{aligned} \tag{9}$$

If $(\hat{\gamma}, \hat{\mu}, \beta, \alpha)$ is optimal and $\alpha > 0$, then $(\hat{\gamma}, \hat{\mu})/\alpha$ is optimal in Eq. (6), with $v = H(\hat{\gamma})$ and $\bar{\mu} = \hat{\mu}$ in Eq. (7).

(b) $F(z)$ is also the optimal value of

$$\begin{aligned} \min_{g, a, \lambda, w, \eta} \quad & \eta \\ \text{subject to} \quad & \ell(\sigma_k g_r) \le a_i^{(k)} + \textstyle\sum_m w_{km} \Delta_{ir}^{(k)}[m], && k = 0,1,\ i \in [n_k],\ r \in [N], \\ & 0 \le w_{km} \le \lambda_k, \quad \rho_0 w_{0m} + \rho_1 w_{1m} \le \eta z_m, && k = 0,1,\ m \in [D], \\ & \textstyle\sum_k \big(\frac{1}{n_k} \sum_i a_i^{(k)} + \rho_k \lambda_k\big) \le \eta, \quad \lambda \in \mathbb{R}^2_+. \end{aligned} \tag{10}$$

(c) If the coupling $\rho_0 w_{0m} + \rho_1 w_{1m} \le \eta z_m$ in Eq. (10) is replaced by the big-M constraints $w_{km} \le (2/\rho_k) z_m$, the optimal value $F_M(z)$ satisfies $F_M \le F$ on $[0,1]^D$ and $F_M(\mathbf{1}_S) = V(S)$.

Appendix B gives the proof. In Eq. (9), $(\hat{\gamma}, \hat{\mu}, \alpha) = (\gamma, \mu, 1)/(1 + \mu^\top z)$, and β linearizes the positive parts. For the hinge loss, $H(\hat{\gamma}) = \max\{2\sum_r h_r : h_r \le p_0^{\hat{\gamma}}(r),\ h_r \le p_1^{\hat{\gamma}}(r)\}$, so Eq. (9) is a linear program. Eq. (10) is the epigraph of Eq. (5) with $\lambda_k z_m$ replaced by w_{km}. Its coupling bounds w_{km} by $\eta z_m/\rho_k$, a bound that scales with the objective. No big-M constant is needed, and by (c) the resulting relaxation is never weaker than the big-M one.

Cuts and master problem. Let $z^t = \mathbf{1}_{S^t}$, let $(\hat{\gamma}^t, \hat{\mu}^t, \beta^t, \alpha_t)$ solve Eq. (9) at z^t, and let $h_t = H(\hat{\gamma}^t)$. By Theorem 3.1(c) and Proposition 4.1(a),

$$c_t(z) = h_t\big(1 - (\hat{\mu}^t)^\top (z - z^t)\big) \tag{11}$$

satisfies $c_t \le F$ on $[0,1]^D$ and $c_t(z^t) = F(z^t) = V(S^t)$. If $\alpha_t = 0$, then $\hat{\gamma}^t = 0$, $h_t = 0$ and $c_t \equiv 0$, which is still valid. Since F is nonincreasing and $d \le D$, some minimizer of F over $\mathcal{Z}_d$ satisfies $\mathbf{1}^\top z = d$. The master problem is therefore

$$L_t = \min_{z \in \{0,1\}^D,\ \eta \ge 0} \eta \quad \text{subject to} \quad \mathbf{1}^\top z = d, \quad \eta \ge c_s(z),\ s = 1, \dots, t, \tag{12}$$

and z^{t+1} denotes a minimizer. Let $\mathrm{OPT} = \min_{|S| \le d} V(S)$. Then $L_t \le \mathrm{OPT} \le U_t = \min_{s \le t} V(S^s)$, because every cut lies below F. Algorithm 1 alternates evaluations and master problems, as in [4] and [1, Algorithm 3.1].

Algorithm 1 Outer approximation for variable selection

Require: Samples $x_i^{(k)}$, radii $\rho_k > 0$, budget $d \le D$, tolerance $\varepsilon > 0$, initial S^1 with $|S^1| = d$

$U \leftarrow +\infty$, $t \leftarrow 1$
loop
  Solve Eq. (5) at S^t for (g^t, a^t, λ^t) and its value $V(S^t)$
  Solve Eq. (9) at $z^t = \mathbf{1}_{S^t}$ and form c_t by Eq. (11)
  if $V(S^t) < U$ **then**
    $U \leftarrow V(S^t)$, $S^\star \leftarrow S^t$, $(g^\star, a^\star, \lambda^\star) \leftarrow (g^t, a^t, \lambda^t)$
  end if
  Solve Eq. (12) with $c_1, \dots, c_t$ for z^{t+1} and L_t
  if $U - L_t \le \varepsilon$ **then**
    return $S^\star$ and the detector $f^\star$ built from $(g^\star, a^\star, \lambda^\star)$ by Eq. (13)
  end if
  $S^{t+1} \leftarrow \{m : z_m^{t+1} = 1\}$, $t \leftarrow t + 1$
end loop

Proposition 4.2 (Finite termination)**.** *If Eq. (5), Eq. (9) and Eq. (12) are solved exactly, Algorithm 1 stops after at most $\binom{D}{d}$ iterations and returns $S^\star$ with $V(S^\star) \le \mathrm{OPT} + \varepsilon$.*

Proof. If $z^{t+1} = z^s$ for some $s \le t$, then $L_t \ge c_s(z^{t+1}) = F(z^s) \ge U_t$, and the algorithm stops. Hence the iterates before termination are distinct points of $\{z \in \{0,1\}^D : \mathbf{1}^\top z = d\}$. On return, $V(S^\star) = U \le L_t + \varepsilon \le \mathrm{OPT} + \varepsilon$. □

Implementation. The code implements the hinge loss. Its scores may then be restricted to $[-1, 1]$ without loss, and Eq. (5), Eq. (9) and Eq. (10) are linear programs. The code also accepts general empirical weights $q_i^{(k)}$ in place of $1/n_k$; nothing above changes. Because solvers return only approximately feasible points, every bound is certified. A solution of Eq. (5) is made feasible by resetting $a_i^{(k)} = \max_r [\ell(\sigma_k g_r) - \lambda_k \mathbf{1}_S^\top \Delta_{ir}^{(k)}]_+$, which gives an upper bound on $V(S^t)$ and a detector. A solution of Eq. (9)

is rescaled to Π, μ is raised until the constraint of Eq. (6) holds, and H is lowered by an explicit error bound. Theorem 3.1(c) then keeps the cut valid, tight or not; it is stored in the form of Remark 3.2. L_t is the master solver's global dual bound, not its incumbent value. The pair Eq. (10) and Eq. (9) evaluates F at fractional z with matching bounds.

Remark 4.3 (Cost and choice of cuts)**.** The $2D$ coupling rows of Eq. (9) contain about DN^2 nonzeros, against about N^2 for Eq. (4), so the cut problems dominate the running time. A tight cut does not require Eq. (9). By the proof of Theorem 3.1(b), a solution $\gamma^\star$ of Eq. (4) at S^t with the prices $\mu^\star$ defined there is optimal in Eq. (6) at $\mathbf{1}_{S^t}$. Then Eq. (7) gives the tight cut $V(S^t)\big(1-(\mu^\star)^\top z\big)$, and Proposition 4.2 still applies. Optimal pairs at binary points are generally not unique, and different choices give different cuts. Selecting Pareto-optimal ones [6] is a natural refinement.

A Proof of Lemma 2.1

Proof. Fix S and let v_S be the infimum in Eq. (5). We show $\max_{\Gamma_S} H \le V(S) \le v_S = \max_{\Gamma_S} H$, and then attainment.

Duality. Let $\Phi(\gamma, g) = \sum_{k,i,r} \gamma_{ir}^{(k)} \ell(\sigma_k g_r)$, so that $H(\gamma) = \inf_g \Phi(\gamma, g)$ by Eq. (3). The set Γ_S is a nonempty polytope, since it contains ι. The function Φ is continuous, linear in γ and convex in g. Sion's theorem [7] therefore gives $\max_{\gamma\in\Gamma_S} H(\gamma) = \inf_g \max_{\gamma\in\Gamma_S} \Phi(\gamma, g)$. The outer maximum is attained because H is upper semicontinuous. For fixed g, the inner maximum splits into one feasible, bounded linear program per class,

$$\max\Big\{\textstyle\sum_{i,r} \gamma_{ir}^{(k)} \ell(\sigma_k g_r) :\ \gamma^{(k)} \ge 0,\ \sum_r \gamma_{ir}^{(k)} = \frac{1}{n_k},\ \sum_{i,r} \gamma_{ir}^{(k)} \mathbf{1}_S^\top \Delta_{ir}^{(k)} \le \rho_k\Big\},$$

whose dual is $\min\{\frac{1}{n_k}\sum_i a_i^{(k)} + \rho_k \lambda_k :\ \lambda_k \ge 0,\ a_i^{(k)} + \lambda_k \mathbf{1}_S^\top \Delta_{ir}^{(k)} \ge \ell(\sigma_k g_r)\ \forall i, r\}$. Adding the two classes and minimizing over g gives $\max_{\Gamma_S} H = v_S$.

Lower bound. Let $\gamma \in \Gamma_S$. Let $y_{ir}^{(k)}$ agree with x_r on S and with $x_i^{(k)}$ off S, and set $\mathbb{P}_k^\gamma = \sum_{i,r} \gamma_{ir}^{(k)} \delta_{y_{ir}^{(k)}}$. Moving mass $\gamma_{ir}^{(k)}$ from $x_i^{(k)}$ to $y_{ir}^{(k)}$ costs $\mathbf{1}_S^\top c_k(\gamma) \le \rho_k$ in total, so $\mathbb{P}_k^\gamma \in \mathcal{P}_k$. Every $f \in \mathcal{F}_S$ satisfies $f(y_{ir}^{(k)}) = f(x_r)$, hence

$$R(f) \ge \sum_{k=0}^{1} \mathbb{E}_{\mathbb{P}_k^\gamma} \ell(\sigma_k f(X)) = \Phi\big(\gamma, (f(x_r))_{r=1}^N\big) \ge H(\gamma).$$

Minimizing over f and maximizing over γ gives $\max_{\Gamma_S} H \le V(S)$.

Upper bound. Let (g, a, λ) be feasible in Eq. (5). If $x_r[S] = x_{r'}[S]$, the constraints on g_r and $g_{r'}$ coincide, so we may take $g_r = g_{r'}$. Then $\hat g(x_r[S]) = g_r$ is well defined on the projected samples. For $\xi \in \mathbb{R}^{|S|}$ and $k = 0, 1$ let

$$E_k(\xi) = \min_{i\in[n_k]} \big\{a_i^{(k)} + \lambda_k \|x_i^{(k)}[S] - \xi\|_1\big\},$$

let i_0, i_1 be the smallest minimizing indices, and write $r_k = \|x_{i_k}^{(k)}[S] - \xi\|_1$ and $b = \|x_{i_0}^{(0)}[S] - x_{i_1}^{(1)}[S]\|_1$. At every ξ that is not a projected sample, set

$$\hat g(\xi) = (1-\tau)\,\hat g(x_{i_0}^{(0)}[S]) + \tau\,\hat g(x_{i_1}^{(1)}[S]), \qquad \tau = \begin{cases} \min\{1, r_0/b\}, & b > 0, \\ 0, & b = 0. \end{cases} \tag{13}$$

We claim that

$$\ell(-\hat g(\xi)) \le E_0(\xi), \qquad \ell(\hat g(\xi)) \le E_1(\xi), \qquad \xi \in \mathbb{R}^{|S|}. \tag{14}$$

At a projected sample $\xi = x_r[S]$ this is the constraint of Eq. (5) at target r, minimized over sources. Otherwise the definition of τ and the triangle inequality $b \le r_0 + r_1$ give $\tau b \le r_0$ and $(1-\tau)b \le r_1$. By convexity of ℓ and the constraints for the sources $x_{i_0}^{(0)}$ and $x_{i_1}^{(1)}$ at these two targets,

$$\ell(-\hat g(\xi)) \le (1-\tau) a_{i_0}^{(0)} + \tau\big(a_{i_0}^{(0)} + \lambda_0 b\big) \le a_{i_0}^{(0)} + \lambda_0 r_0 = E_0(\xi),$$
$$\ell(\hat g(\xi)) \le (1-\tau)\big(a_{i_1}^{(1)} + \lambda_1 b\big) + \tau a_{i_1}^{(1)} \le a_{i_1}^{(1)} + \lambda_1 r_1 = E_1(\xi).$$

The anchors are chosen by finitely many comparisons of continuous functions, so $\hat{g}$ is Borel measurable, and $|\hat{g}| \le \max_r |g_r|$. Let $f(x) = \hat{g}(x[S])$, let $\mathbb{P}_k \in \mathcal{P}_k$ and $\epsilon > 0$, and take a coupling of $\widehat{\mathbb{P}}_k$ and $\mathbb{P}_k$ with cost at most $\rho_k + \epsilon$, written as $\frac{1}{n_k}\sum_i \delta_{x_i^{(k)}} \otimes \pi_i$. By Eq. (14), $\ell(\sigma_k f(x)) \le a_i^{(k)} + \lambda_k \|x_i^{(k)} - x\|_1$ for every i, so

$$\mathbb{E}_{\mathbb{P}_k} \ell(\sigma_k f(X)) = \frac{1}{n_k}\sum_{i=1}^{n_k} \int \ell(\sigma_k f(x))\, d\pi_i(x) \le \frac{1}{n_k}\sum_{i=1}^{n_k} a_i^{(k)} + \lambda_k(\rho_k + \epsilon).$$

Letting $\epsilon \to 0$ and adding the classes gives that $R(f)$ is at most the objective of (g, a, λ). Hence $V(S) \le v_S$.

Attainment. The point $g = 0$, $a = \mathbf{1}$, $\lambda = 0$ is feasible. The constraint whose target is the source itself gives $a_i^{(k)} \ge 0$. On the set where the objective is at most B, therefore, $0 \le a_i^{(k)} \le n_k B$ and $0 \le \lambda_k \le B/\rho_k$, so every $\ell(g_r)$ and $\ell(-g_r)$ is bounded. Being convex, nondecreasing and nonconstant, ℓ satisfies $\ell(t) \to \infty$ as $t \to \infty$, so g is bounded as well. Thus Eq. (5) attains its minimum. For a minimizer, $R(f) \le v_S = V(S) \le R(f)$, so f attains $V(S)$. □

B Proof of Proposition 4.1

Proof. (a) Let (γ, μ) be feasible in Eq. (6) and put $\alpha = 1/(1 + \mu^\top z)$, $(\hat{\gamma}, \hat{\mu}) = \alpha(\gamma, \mu)$ and $\beta_{km} = \alpha(c_k(\gamma)[m] - \rho_k \mu_m)_+$. As c_k and H are positively homogeneous, this point is feasible in Eq. (9) with value $H(\hat{\gamma}) = H(\gamma)/(1 + \mu^\top z)$. Conversely, let $(\hat{\gamma}, \hat{\mu}, \beta, \alpha)$ be feasible in Eq. (9). If $\alpha = 0$, the row constraints force $\hat{\gamma} = 0$, and the value is $0 \le F(z)$. If $\alpha > 0$, then $\beta_{km} \ge (c_k(\hat{\gamma})[m] - \rho_k \hat{\mu}_m)_+$ shows that $(\gamma, \mu) = (\hat{\gamma}, \hat{\mu})/\alpha$ is feasible in Eq. (6). Moreover $1 + \mu^\top z = 1/\alpha$, so its ratio is $H(\hat{\gamma})$ and $\bar{\mu} = \hat{\mu}$. Both programs therefore have the same value, and optimal solutions correspond.

(b) Attach multipliers $\gamma \ge 0$ to the loss constraints, $\beta \ge 0$ to $w \le \lambda$, $\mu \ge 0$ to the coupling and $\alpha \ge 0$ to the epigraph constraint. The Lagrangian is

$$\begin{aligned}\mathcal{L} = \eta(1 - \alpha - \mu^\top z) + \sum_{k,i,r} \gamma_{ir}^{(k)} \ell(\sigma_k g_r) + \sum_{k,i} a_i^{(k)} \Big(\frac{\alpha}{n_k} - \sum_r \gamma_{ir}^{(k)}\Big) \\ + \sum_k \lambda_k \Big(\rho_k \alpha - \sum_m \beta_{km}\Big) + \sum_{k,m} w_{km}\big(\beta_{km} + \rho_k \mu_m - c_k(\gamma)[m]\big).\end{aligned}$$

Its infimum over g is $H(\gamma)$ by Eq. (3). Its infimum over the free variables η, a and the nonnegative variables λ, w is finite exactly when the constraints of Eq. (9) hold. Hence the Lagrangian dual of Eq. (10) is Eq. (9). The program Eq. (10) is convex. Its only nonaffine constraints are the loss constraints, and they hold strictly at $g = 0$, $a = 2 \cdot \mathbf{1}$, $\lambda = 0$, $w = 0$, $\eta = 4$, which satisfies the affine constraints. Its value is finite, because the constraints whose target is the source give $a \ge 0$ and hence $\eta \ge 0$. Slater's condition [2, §5.2.3] gives equal values, which equal $F(z)$ by (a).

(c) As in Appendix A, the sublevel sets of Eq. (10) are compact, now with $0 \le w_{km} \le \lambda_k \le \eta/\rho_k$. Hence a minimizer exists, and $\eta = F(z) \le 2$ there. It satisfies $\rho_k w_{km} \le \eta z_m \le 2 z_m$, so it is feasible in the big-M program with objective at most $F(z)$, and $F_M(z) \le F(z)$. At $z = \mathbf{1}_S$ the big-M constraints force $w_{km} = 0$ for $m \notin S$, and $w_{km} \le \lambda_k$ gives $\sum_m w_{km} \Delta_{ir}^{(k)}[m] \le \lambda_k \mathbf{1}_S^\top \Delta_{ir}^{(k)}$. Every feasible point therefore yields a feasible point of Eq. (5) with the same objective, and $F_M(\mathbf{1}_S) \ge V(S) = F(\mathbf{1}_S)$. □

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